

Electric Conduction – Solved Exercises

1. Using the Richardson–Dushman equation

$$J = AT^2 \exp\left(-\frac{W}{kT}\right),$$

the ratio of the emitted currents at temperatures $T_1 = 2200$ K and $T_2 = 2350$ K is

$$\frac{J_2}{J_1} = \left(\frac{T_2}{T_1}\right)^2 \exp\left[-W\left(\frac{1}{kT_2} - \frac{1}{kT_1}\right)\right].$$

Substituting $W = 4.5$ eV and $k = 8.617 \times 10^{-5}$ eV/K yields

$$\frac{J_2}{J_1} \approx 5.19.$$

2. The retarding voltage reduces the emission current density as

$$J = J_0 \exp\left(-\frac{|q|V}{kT}\right),$$

and we require

$$\frac{J}{J_0} = \frac{1}{15}.$$

Thus

$$V = \frac{kT}{|q|} \ln 15.$$

For $T = 2200$ K,

$$V \approx 0.045 \text{ V}.$$

3. The drift velocity in a copper wire is given by

$$v_d = \frac{I}{nqA},$$

where $A = \pi R^2$ with $R = 1.5$ mm and $n = 3.3 \times 10^{28} \text{ m}^{-3}$. Thus,

$$v_d = \frac{12}{(3.3 \times 10^{28})(1.6 \times 10^{-19})\pi(1.5 \times 10^{-3})^2} \approx 2.04 \times 10^{-4} \text{ m/s}.$$

4. We examine an n-type semiconductor under the conditions:

$$B_y = 0.4 \text{ T}, \quad U_H = 10 \text{ mV}, \quad I = 30 \text{ mA}.$$

- (a) The Hall effect gives

$$U_H = \frac{IB}{nqt},$$

hence the electron concentration

$$n = \frac{IB}{|q|U_H t}.$$

(The thickness t must be known to compute a numerical value.)

(b) The drift velocity is

$$v_d = \frac{I}{nqA},$$

where A is the cross-sectional area of the sample.

(c) The Hall constant is

$$R_H = \frac{1}{nq}.$$